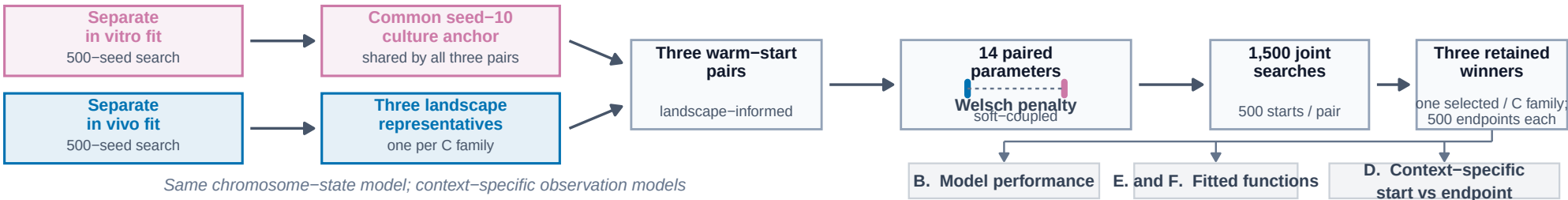


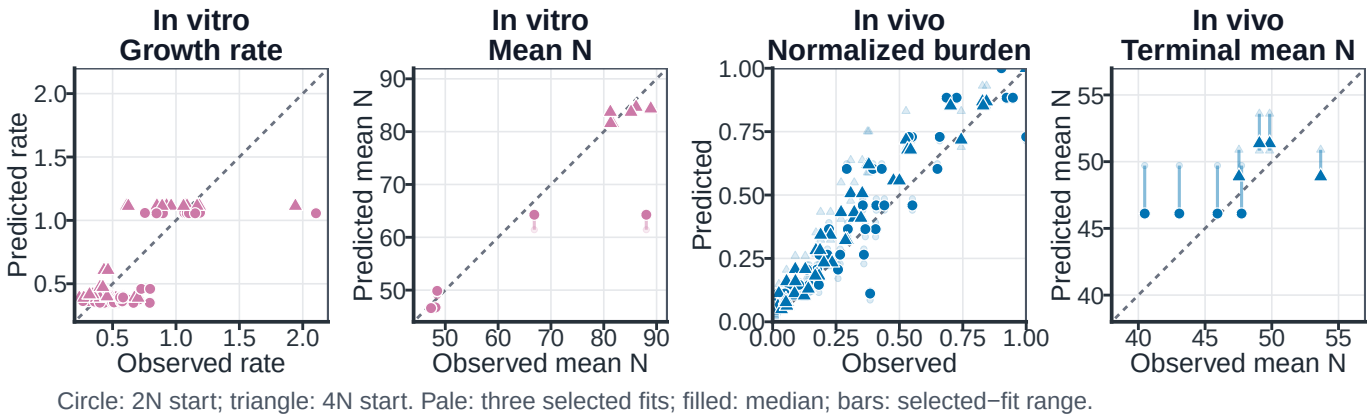
Joint-fit workflow, performance, mechanisms, and solution stability

Three primary-family pairs; Panel D compares pair-specific DE initial populations with 500 optimizer endpoints.

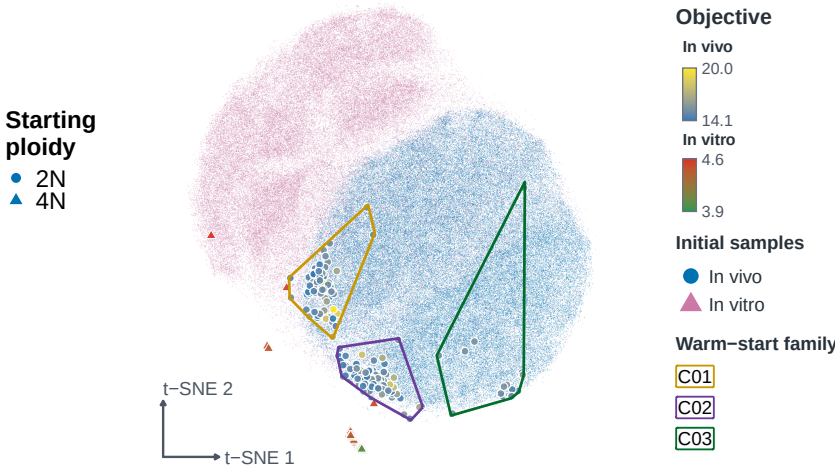
A. Joint-fitting workflow



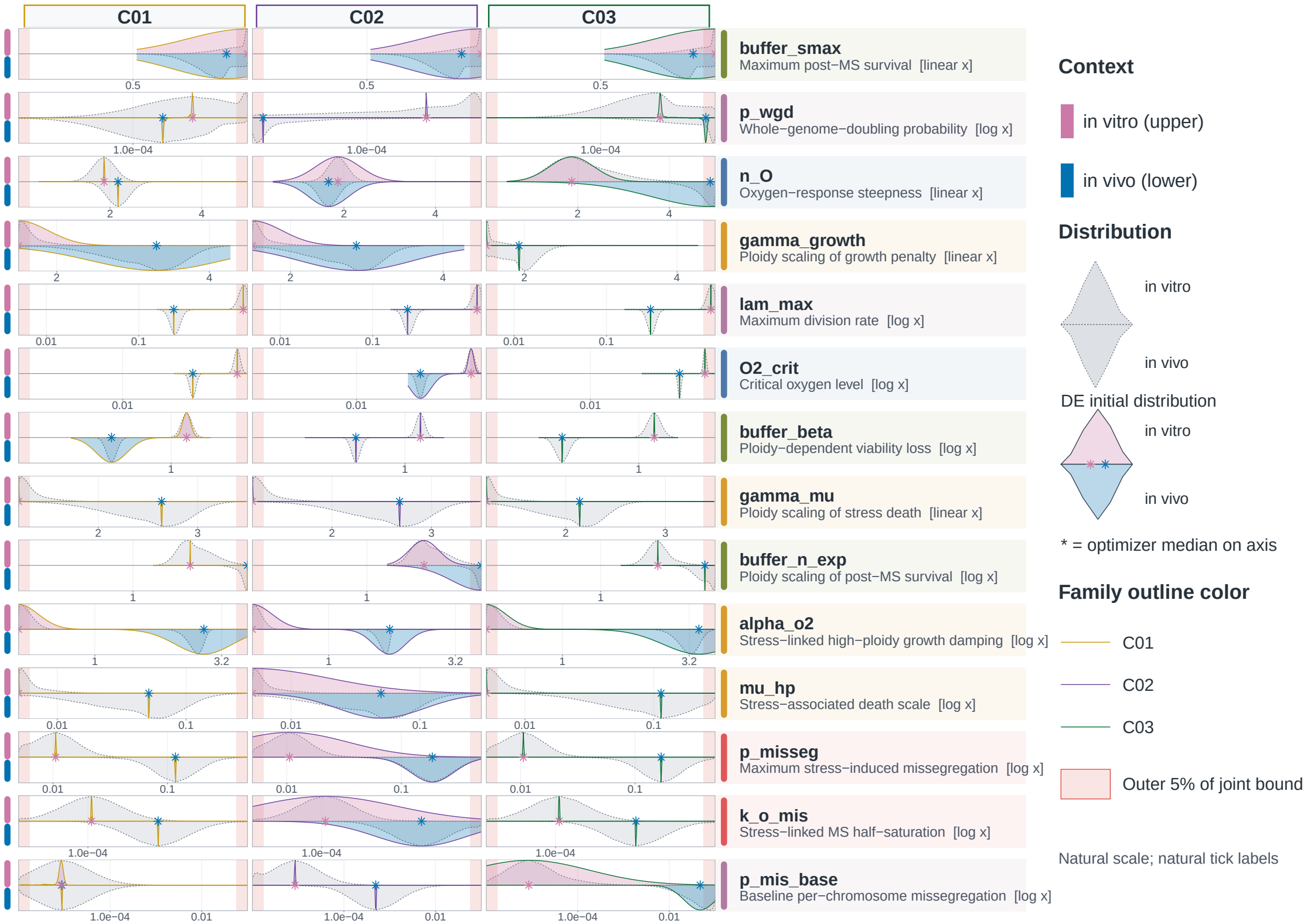
B. In-sample model performance



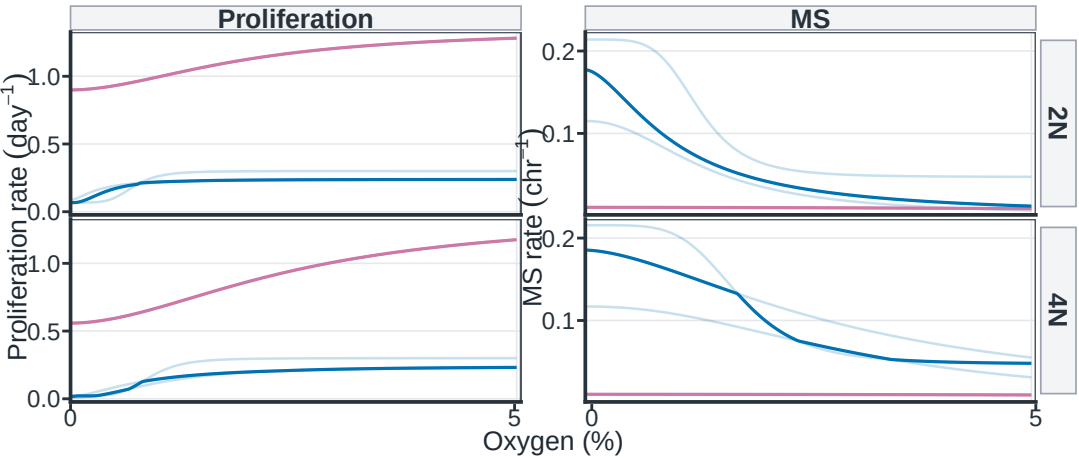
C. Warm-start map



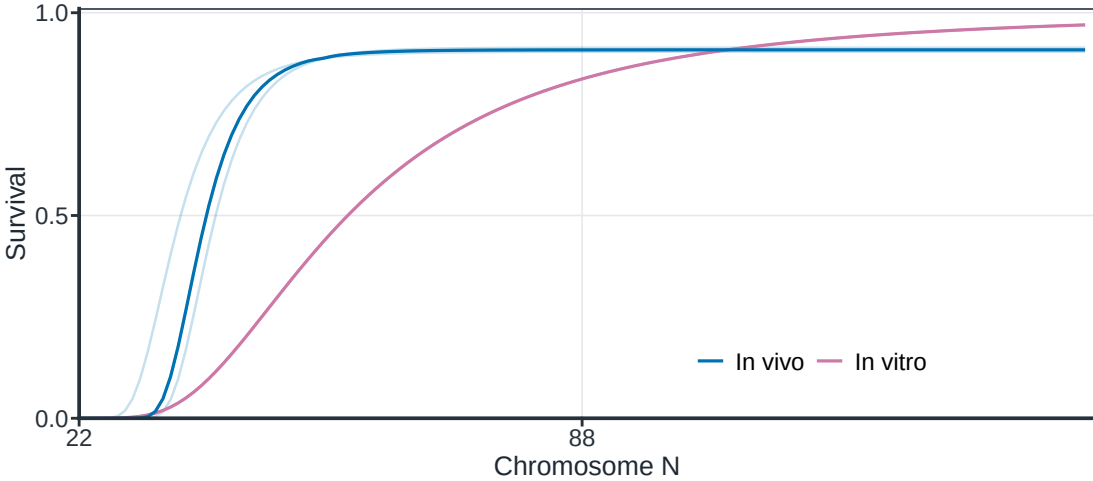
D. Context-specific DE initialization and optimizer endpoints



E. Oxygen-dependent fitted functions



F. Survival



B: circle = 2N start; triangle = 4N start. E-F: thin curves = three fits; thick curves = pointwise medians.
D: upper = in vitro; lower = in vivo; gray fill/dashes = DE initialization; pink/blue fill with C-family solid outlines = optimizer endpoints.
Red background = outer 5% of the complete joint bound.
C01/C02/C03 retain the same family identities across panels; optimizer distributions are descriptive numerical-search summaries, not posterior or confidence distributions.